Phase 1: Data Ingestion (Weeks 1–2)

*These three tracks can and should be done in **parallel**.*

Track A: Financial & Market Data

- **Step 1:** Scrape MSRB EMMA for coastal NJ bond issuances (2015-2025). Extract Issuer, Sale Date, Maturity, Yield, Coupon, Rating, Insurance status, and Tax Status.
- **Step 2:** Pull historical AAA MMD Benchmarks from the Bloomberg Terminal.
- **Step 3:** Calculate the Spread over AAA for every bond in the dataset.

Track B: Spatial & Climate Data

- **Step 4:** Download the NJ PACT 2.1ft, 4.1ft, and 6.1ft SLR spatial layers (Shapefiles/GeoJSON).
- **Step 5:** Download NJ Municipal Boundaries.
- **Step 6:** Perform a spatial join (intersection) to calculate the exact percentage of each municipality's area/tax base that falls within those SLR zones.

Track C: Resilience & Control Data

- **Step 7:** Scrape/clean the FEMA CRS Class data for NJ municipalities.
- **Step 8:** Process the internal I-Bank priority lists (once received from your NJDEP contact) to flag the "150-point" resilient projects.
- **Step 9:** Download NJ DCA municipal budget data (Debt-to-GDP) and Census ACS data (Median Household Income).

Phase 2: Data Integration & Entity Resolution (Week 3)

Dependency: Must be completed after Phase 1 is finished.

- **Step 10:** Standardize municipality names. (EMMA might say "Sea Isle City", Census says "Sea Isle City city", FEMA says "CITY OF SEA ISLE CITY"). Use a fuzzy string-matching script to align all datasets.
- **Step 11:** Merge Tracks A, B, and C into a single, unified analytical panel dataset (CSV/SQL database).
- **Step 12:** Classify the cohorts. Create the binary/categorical variables for Resilient vs Non-Resilient, and High-Exposure vs Low-Exposure.

Phase 3: Econometric Modeling (Weeks 4–6)

Dependency: Relies entirely on the clean dataset from Phase 2.

- **Step 13:** Run the Matched-Pair Analysis. Compare high-exposure/high-resilience towns directly against high-exposure/low-resilience towns with the same credit rating.
- **Step 14:** Run the Difference-in-Differences (DiD) regression focused on the "2022 Pivot." Measure if spreads compressed for I-Bank funded towns after the 150-point rule was introduced.
- **Step 15:** Extract the final basis point (bps) coefficients for the "SLR Penalty" and the "Resilience Premium."

Phase 4: Web Dashboard Development (Weeks 6–8)

Parallel Opportunity: The frontend wireframing (Steps 16-17) can begin during Phase 3, but the data binding (Step 18) depends on the econometric outputs.

- **Step 16 (Architecture setup):** Set up the application infrastructure. A React frontend paired with a FastAPI backend will give you the snappiest performance for rendering complex interactive maps and serving the econometric data dynamically.
- **Step 17 (Frontend UI):** Build the two primary views:
  - *Macro View:* Top-level metrics displaying the state-wide average Resilience Premium (e.g., "Average Savings: 15 bps").
  - *Micro View:* An interactive Mapbox or Leaflet choropleth map of New Jersey.
- **Step 18 (Backend Data Binding):** Connect the model outputs to the frontend.
  - When a user clicks on a specific municipality polygon, trigger a pop-up card displaying its specific stats: CRS Score, % Tax Base at Risk (2.1ft/4.1ft/6.1ft), Total Bond Debt, and Estimated $ Interest Saved/Lost.

Phase 5: Synthesis & Handoff (Week 9)

Dependency: All previous phases must be complete.

- **Step 19:** Draft the formal Technical Paper explaining the methodology, controls, and limitations.
- **Step 20:** Finalize the slide deck tailored for state and local planners, heavily featuring screenshots from the dashboard to prove ROI.
- **Step 21:** Deploy the web dashboard to a cloud host (AWS/Heroku/Vercel) so your NJDEP contact has a live URL to share internally.